Hailan Shanbhag

Lausanne, CH hailanshanbhag@gmail.com +41 76 263 51 78

Citizenship: USA on Permit B (Switzerland) hailanshanbhag.github.io hailan-shanbhag hailanzs

Education

École Polytechnique Fédérale de Lausanne *Jan 2023 – present*
PhD in Computer and Communication Sciences

University of Illinois Urbana Champaign *Aug 2021 – Dec 2022*
MS in Electrical and Computer Engineering

- GPA: 4.0/4.0

University of Illinois Urbana Champaign *Aug 2017 – May 2021*
BS in Computer Engineering with High Honors

- GPA: 3.84/4.0

Publications

[CVPR '26] "Seeing through boxes: Non-Line-of-Sight 3D Reconstruction from Radar Signals". *The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026.*

Jiachen Lu* , ***Hailan Shanbhag****, Haitham Hassanieh

[NeurIPS '25] "GeRaF: Neural Geometry Reconstruction from Radio Frequency Signals". *The Thirty-Ninth Annual Conference on Neural Information Processing Systems.* ***(Spotlight)***

Jiachen Lu* , ***Hailan Shanbhag****, Haitham Hassanieh

[MobiCom '24] "Around the Corner mmWave Imaging in Practical Environments". *In ACM International Conference on Mobile Computing and Networking.*

Laura Dodds, ***Hailan Shanbhag***, Saurabh Gupta, Haitham Hassanieh

[MobiSys '23] "Contactless Material Identification with Millimeter Wave Vibrometry". *In ACM International Conference on Mobile Systems, Applications, and Services.*

Hailan Shanbhag*, Sohrab Madani*, Akhil Isanaka, Deepak Nair, Saurabh Gupta, Haitham Hassanieh

(* indicates equal contribution, co-primary first authors)

Experience

Doctoral Research Assistant *Lausanne, Champaign*
EPFL, Prof. Haitham Hassanieh *Jan 2023 – present*
UIUC, Prof. Haitham Hassanieh *Aug 2021 – Dec 2022*

- Designed and implemented a wireless material sensing system using both acoustics and mmWave signals.
- Implemented novel algorithms for non-line-of-sight millimeter-wave imaging in practical scenarios.
- Studied 3D neural reconstruction methods of complex objects in non-line-of-sight using synthetic aperture radar and robotics.

Senior Research Project & Thesis *Champaign, IL*
UIUC, Prof. Haitham Hassanieh *Aug 2020 – May 2021*

- Calibrated four 60 GHz Qualcomm phased array antennas to construct a 12x12 MIMO array for both a transmitter and receiver (hardware acquired from the M-Cube project of UCSD).
- Measured beam patterns of the transmitter and receiver radios and prepared the hardware for future applications.
- Implemented a fast calibration algorithm for antenna calibration on the 12x12 MIMO array.

Silicon Verification Intern *Sunnyvale, CA*
Microsoft *Jun 2019 – Aug 2019*

- Enhanced a UVM based verification IP by providing support for OCP VIPs.

- Created a translation layer from the AXI protocol to the OCP protocol, which was integrated into an inhouse verification IP.
- Integrated part of the translation layer via fully synthesizable code to reuse an inhouse AXI slave.

Undergraduate Research Assistant *Champaign, IL*
UIUC, Prof. Viktor Gruev *May 2018 – May 2019*

- Designed and fabricated a PCB for a Hamamatsu CMOS area image sensor realizing low-noise multi-spectral imaging for image-guided surgery and underwater polarization imaging.
- Began programming XEM7310 OpalKelly FPGA in Verilog to communicate with and process LVDS pixel data received from the image sensor.
- Communicated to the FPGA using OpalKelly's FrontPanel C++ API to interface through a PC.

Teaching

Mobile Networks (COM 405)
Fall 2024, Fall 2025

- Designed, tested and ran a hardware lab students to have a hands on exercise with TI's mmWave radars for ranging, real-time vital sign monitoring and non-line-of-sight imaging.
- Created radar imaging lab exercises for image processing algorithms.
- Held office hours and guided students through material on all things wireless.

Intelligent Systems: Communications & AI (COM 304)
Fall 2023, Spring 2024, Spring 2025, Spring 2026

- Helped create and teach a project-based course on robotics, wireless communications, sensing and AI.
- Filmed video lectures on wireless systems and signal processing.
- Prepared tutorials and created a Python codebase for TI's AWR1843 77 GHz radar.
- Met with students weekly to help with problems and guide their projects (e.g. Radar SLAM, real-time human tracking).

Talks

CVPR Workshop Presenter: Bringing Computer Vision to Hidden Objects
CVPR, June 2026

Conference Presentation: Contactless Material Identification with Millimeter Wave Vibrometry
MobiSys, June 2023

Fellowships & Awards

IC School Excellent Teaching Assistant Award	*December 2025*
Promise of Excellence Fellowship	*Aug 2021 – May 2022*
TI Women STEM Stars Scholarship	*Aug 2017 – May 2021*

Skills

Languages: Python, MATLAB, C++/C, Verilog/SystemVerilog, x86

Frameworks: PyTorch, Eagle, KiCad, mmWave Studio, CUDA, ROS

Hardware: TI 77 GHz Radar (XWR1843), Franka Research 3

Spoken Languages: English (native), Mandarin Chinese (first language), French (B1), German (A1)

Service

NeurIPs Reviewer	*2026*
MobiSys Artifact Evaluation Committee Member	*March 2026*
CVPR Reviewer	*2026*
EDIC Pre-Filtering Committee	*December 2024*

MobiCom Student Volunteer *March 2022*

Secondary Reviewer for MobiCom, SIGCOMM, SenSys and MobiSys *January 2022 - present*

Supervised Students

Eugen Bosnjak: EPFL Master Project, Spring 2025 - Winter 2025

Andrea Tarabay: EPFL Summer Intern, Summer 2024

Charchit Gupta: EPFL Summer Intern, Summer 2023

Deepak Nair: UIUC Summer Intern, Summer 2022

Akhil Isanaka: UIUC Summer Intern, Summer 2022